1. *Without* using a calculator, find the value of each of the following.

(a) $49^{\frac{1}{2}}$ (b) $27^{\frac{1}{3}}$ (c) $16^{\frac{1}{4}}$ (d) $8^{\frac{1}{3}}$

(e) $100^{\frac{1}{2}}$ (f) $9^{1.5}$ (g) $16^{\frac{3}{4}}$ (h) $125^{\frac{2}{3}}$

(i) $32^{\frac{4}{5}}$ (j) $36^{\frac{1}{2}}$ (k) $64^{\frac{1}{2}}$ (l) $121^{\frac{1}{2}}$

(m) $27^{\frac{2}{3}}$ (n) $81^{\frac{3}{4}}$ (o) $1000^{\frac{2}{3}}$ (p) $32^{0.6}$

(q) $4^{1.5}$ (r) $4^{2.5}$ (s) $81^{0.25}$ (t) 5^{-1}

(u) $\left(\frac{1}{4}\right)^{-2}$ (v) $8^{-\frac{1}{3}}$ (w) $9^{-0.5}$ (x) $16^{-0.25}$

(y) $32^{-0.8}$ (z) $16^{-1.5}$

2. What is the value of each of the following expressions? *Do not use a calculator.*

(a) $\left(\frac{1}{4}\right)^{\frac{1}{2}}$ (b) $\left(\frac{1}{8}\right)^{-\frac{2}{3}}$ (c) $(25)^{-2.5}$

(d) $\left(\frac{1}{16}\right)^{-\frac{1}{4}}$ (e) 7^0 (f) $\left(\frac{4}{5}\right)^{-2}$

(g) $\left(\frac{4}{7}\right)^{-2}$ (h) $5^{-2} \times 4^3$ (i) $2^{-3} \div 3^{-3}$

(j) $3^4 \div \left(\frac{1}{3}\right)^{-4}$ (k) $\left(\frac{4}{9}\right)^{-2} \times \left(\frac{27}{8}\right)^{-3}$ (l) $78^{-1} \times 13^3$

3. Express each of the following in index form.

(a) $\left(a^{16}\right)^{\frac{1}{2}}$ (b) $\left(a^{15}\right)^{\frac{1}{3}}$ (c) $\left(27a^{9}\right)^{\frac{1}{3}}$ (d) $\left(x^{32}\right)^{\frac{1}{4}}$

(e) $\left(x^{20}\right)^{\frac{1}{4}}$ (f) $a^{-2} \div a^{-4}$ (g) $a^{4} \times a^{-5}$ (h) $\left(a^{2}\right)^{-\frac{1}{2}}$

(i) $\left(a^{-4}\right)^{2}$ (j) $a^{-4} \times a^{4}$ (k) $\left(a^{\frac{1}{2}}\right)^{4}$ (l) $\left(a^{16}\right)^{-\frac{1}{4}}$

4. Solve the following equations for x.

(a) $3^{x} = 81$ (b) $4^{x} = 64$ (c) $5^{x} = 125$

(d) $7^{x} = 49$ (e) $2^{x} = 32$ (f) $2^{x} = 64$

(g) $x^{3} = 27$ (h) $x^{3} = 64$ (i) $x^{5} = 32$

(j) $5x^{2} = 45$ (k) $3x^{3} = 24$ (l) $4x^{4} = 324$

(m) $5x^{3} = 320$ (n) $x^{\frac{1}{2}} = 3$ (o) $x^{-4} = 256$

(p) $x^{-\frac{1}{2}} = 5$ (q) $2^{x} = 1$ (r) $3^{-x} = 1$

5. Simplify the following:

a) $\left(\frac{16a^{4}}{b^{4}}\right)^{\frac{3}{4}}$ b) $\left(\frac{a^{2}}{25b^{3}}\right)^{-\frac{3}{2}}$

c) $\frac{12x^{3}\sqrt{x}}{4x^{2}}$ d) $\frac{\sqrt{4z}}{z^{3}}$

e) $\frac{12a^{14}b^{8}}{(3a^{4}b^{3})\times(2a^{2}b^{3})}$ f) $\frac{(4a^{2}b^{3})^{2}}{\sqrt{64a^{2}}}$

g) $(16w^{20})^{\frac{1}{4}}$ h) $(32q^{10})^{\frac{1}{5}}$

i) $\frac{(x^{2})^{-3}}{x^{3}\times x^{-1}}$ j) $\frac{(4a^{-3})^{3}}{2a^{4}\times 4a^{2}}$

6. Solve the following equations:

a) $512^x = \frac{1}{2}$

b) $5^{\frac{4x}{3}} = 125^{\frac{2}{3}}$

c) $\frac{\sqrt{m}}{m^x} = m^2$

d) $\frac{n^3}{\sqrt{n^x}} = \frac{1}{n^2}$

e) $\frac{m^{x-3}}{m^4} = \frac{1}{\sqrt{m}}$

f) $\left(\frac{3}{5}\right)^x = \frac{27}{125}$

g) $\left(\frac{9}{4}\right)^x = \frac{32}{243}$

h) $9^{2x-1} = 27^{x+2}$

i) $32^{3x-1} = 16^{4x-3}$

j) $\frac{8^x \times 64^3}{16^7} = 32$

k) $\frac{81^x \times 9^2}{27^4} = 1$